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Design and Development of an Integrated Study Hub Management System for WS Students & Professionals Lounge

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Abstract

The rapid expansion of shared workspaces has increased the operational demand for efficient facility management systems. The WS Students & Professionals Lounge – La Paz Branch previously managed its daily transactions through manual workflows, paper logbooks, decentralized spreadsheets, and third-party social media platforms. This fragmented operational setup resulted in structural vulnerabilities, including resource double-bookings, prolonged customer waiting times, data entry errors, and financial tracking discrepancies. To mitigate these administrative bottlenecks, this study developed the Integrated Study Hub Management System, a centralized, cross-platform web application designed to optimize back-office utilities and enhance user service quality. Utilizing an agile development methodology, the system was engineered with six primary modules: Reservation Management, Membership and Time Tracking, Payment Billing, Sales Recording and Reporting, Admin and Staff Dashboards, and Role-Based Access Control (RBAC).

The technical architecture was built on a web-based framework supported by a relational MySQL database engine, ensuring secure data synchronization and eliminating data fragmentation. Evaluation metrics indicated that the automation of elapsed-time calculations and live space allocation rules effectively eradicated transcription errors, secured historical financial ledgers, and minimized context-switching for frontline desk personnel. Ultimately, the integration of this system converted manual front-desk friction into actionable business intelligence, providing a secure, transparent, and frictionless facility environment for both administrative personnel and clients.

CHAPTER I

INTRODUCTION

1.1 INTRODUCTION TO THE STUDY

In the modern digital era, the operational efficiency of service-oriented enterprises relies heavily on the integration of centralized software architectures that mitigate manual friction and eliminate data fragmentation. Shared workspace facilities, such as study hubs and co-working lounges, process complex daily transaction streams that demand high computational precision. These transactions include real-time space scheduling, user subscription authentication, and dynamic, time-based financial billing. When an enterprise manages these highly volatile parameters using non-integrated, manual procedures, its operational framework becomes fundamentally vulnerable to processing delays, human error, and structural data silos. Consequently, modern organizational management dictates that transitioning to an automated, centralized platform is an operational necessity to preserve data integrity and optimize service delivery.

The WS Students & Professionals Lounge – La Paz Branch previously conducted its business functions through a fragmented combination of manual, paper-reliant processes, localized spreadsheet applications, and third-party social media utilities. Specifically, the establishment utilized Facebook and Facebook Messenger as its primary communication interfaces for coordinating customer reservations and handling client availability inquiries. Frontline desk personnel had to manually parse incoming chat logs, verify physical availability, and retrospectively transcribe these transactional details into decentralized paper ledgers or detached spreadsheets. This disconnected manual workflow demanded significant administrative labor and created a highly volatile operational environment where data synchronization could not be guaranteed in real time. Additionally, the lounge's customer registration and long-term membership management followed an identical manual methodology. New clients were registered via handwritten forms, and the facility issued physical membership cards while administrative staff manually logged subscription dates into individual spreadsheets. This management model created extreme operational friction when tracking, updating, or verifying expiration timelines as the lounge's active user base expanded. Because there was no

centralized database engine to link these operations, front-desk personnel were constantly forced to context-switch between digital chat windows, physical logbooks, and desktop spreadsheet rows simply to execute a single check-in or validate a membership tier. This procedural friction rendered daily front-office operations heavily time-consuming and highly susceptible to data entry errors.

As a result, the La Paz branch faced persistent operational liabilities that threatened its business performance and customer satisfaction metrics. These liabilities manifested as accidental room double-bookings, overlooked or missed client reservations, asynchronous data updates, and prolonged queues at the service counter. Managing overlapping transaction records across disconnected media significantly inflated the administrative workload of staff, leading to severe service bottlenecks during high-volume operating hours. These systemic inefficiencies collectively highlighted an urgent, immediate requirement for a structured, unified, and permanent technological intervention.

To address these operational bottlenecks, this study developed the Integrated Study Hub Management System for WS Students & Professionals Lounge to centralize and automate key operations such as user registration, room reservations, membership tracking, billing, time-duration tracking, and automated sales reporting. By replacing manual processes with a digital solution, the system successfully reduced administrative errors, streamlined facility workflows, and improved overall service efficiency. Ultimately, the system was designed and implemented to simplify administrative procedures for staff while providing a faster, seamless, and more convenient experience for customers.

1.2 Background of the Study

study hubs and co-working spaces have experienced an unprecedented surge in popularity. This growth is largely driven by the increasing demand from students, freelancers, and working professionals for structured, quiet, and highly productive environments that foster deep focus and collaboration. These spaces fulfill a critical societal and infrastructure need by providing essential facilities, including high-speed internet connectivity, ergonomically sound work areas, proper lighting, and specialized meeting rooms. However, as customer volume expands, the operational complexity of managing these shared facilities scales accordingly. To sustain high service quality and maintain customer satisfaction, modern study hubs are increasingly required to manage their day-to-day administrative frameworks with a high degree of operational efficiency.

The WS Students & Professionals Lounge – La Paz Branch is one such study hub that offers dedicated spaces for studying, meetings, and collaboration. Previously, however, the establishment relied almost exclusively on a combination of fragmented, manual procedures, handwritten logbooks, and legacy spreadsheet software to administer its daily operations. Furthermore, the lounge utilized external social media channels— primarily Facebook and Facebook Messenger—as its primary interfaces for handling client inquiries and booking requests. This operational framework spanned across multiple administrative silos, including customer registration, workspace reservations, membership subscription tracking, elapsed time logging, invoicing, and retrospective financial reporting. Because these workflows were fundamentally decentralized and detached from one another, data retrieval and verification processes remained heavily disorganized and administratively cumbersome

This reliance on non-integrated platforms exposed the establishment to significant operational vulnerabilities. Frontline personnel had to constantly navigate and manually cross-reference communication threads on social media, physical logbooks, and individual digital spreadsheets simply to verify a single customer transaction or room availability status. This structural fragmentation frequently resulted in serious service bottlenecks, such as accidental resource double-bookings, delayed response times to online clients, prolonged front-desk waiting lines, and chronic inaccuracies in tracking real-time room usage and membership status. The logistical strain was particularly magnified during peak operating hours when the high influx of clients drastically

increased the administrative workload, rendering the previous workflow acutely susceptible to human calculation and transcription errors. To resolve these documented administrative bottlenecks, there existed a critical and immediate imperative to transition from a manual, paper-reliant architecture to a secure, digitized infrastructure. The Integrated Study Hub Management System for WS Students & 3 Professionals Lounge was directly engineered to address these operational vulnerabilities. By consolidating and automating the core business mechanisms of client registration, interactive reservations, membership management, time-duration tracking, instant billing computation, and real-time sales reporting into a centralized web-based platform, the system established an integrated, database-driven workflow. Ultimately, the system was designed and successfully implemented to improve operational efficiency, reduce human error, and enhance overall service quality for both staff and customers.

1.3 Statement of the Problem

As the demand for study hubs and co-working spaces continues to grow, businesses like WS Students & Professionals Lounge faced significant operational challenges due to their previous reliance on manual and paper-based management systems. The lack of a centralized platform for managing reservations, memberships, payments, and sales often led to double bookings, long waiting times, and inaccuracies in tracking room usage and sales reports. Because most records were maintained in physical logs or scattered spreadsheets, the business was highly prone to data loss, human error, and delays in service delivery.

Without an integrated system, staff were forced to spend excessive time manually verifying records and updating information, which resulted in operational inefficiency and customer dissatisfaction, especially during peak hours. Furthermore, the absence of real-time reporting made it difficult for management to monitor sales and occupancy accurately, hindering their ability to make data-driven decisions. To maintain its reputation and provide quality service, it was essential for the lounge to transition into an automated and secure management framework.

Specifically, this study sought to address and evaluate the following key objectives and questions:

1. How was the real-time reservation module designed and implemented to effectively prevent double bookings, optimize table allocation, and minimize client waiting times?

2. In what ways was the membership tracking system developed to automatically monitor user activity, track usage duration, and manage subscription validity accurately?

3. What methods were applied to integrate billing and sales reporting so that every transaction was automatically recorded and updated to minimize manual errors?

4. What approaches were employed to implement a centralized dashboard that displayed real-time information on time usage, occupancy, and sales to assist management in decision-making?

5. How did the web-based management system improve the overall service quality and customer satisfaction of WS Students & Professionals Lounge compared to its previous manual processes?

6. What technical features were implemented to ensure that data and transaction records were securely stored and accessible only to authorized personnel?

7. In what ways did the system improve the customer experience in terms of faster transactions, reliable booking, and efficient service delivery?

8. To what extent did the Integrated Study Hub Management System enhance the overall management of user registrations and memberships?

9. How effective was the system in handling automated billing, time-based room usage, and subscription expiration tracking?

10. What impact did the system have on the accuracy and efficiency of automated sales report generation?

1.4 Objectives of the Study

The primary objective of this study was to design, develop, and evaluate an Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch. The system aimed to enhance operational efficiency by automating and streamlining core business processes, improving the quality and speed of service delivery, and ensuring accurate, reliable, and centralized management of user registration, membership records, billing transactions, time-based room usage, and sales report generation.

Specifically, this study aimed to:

1. Design and develop a centralized web-based management system that served as a single platform for all lounge operations and data storage.
2. Automate the reservation and booking process to effectively eliminate double bookings and optimize table allocation for customers.
3. Implement an automated membership and time usage tracking feature to monitor customer subscriptions and facility usage duration accurately.
4. Develop an integrated billing and sales recording system that minimized human errors and ensured all financial transactions were logged correctly.
5. Generate comprehensive real-time reports for sales, room occupancy, and time tracking to assist management in data-driven decision-making.
6. Conduct system testing and evaluation to ensure that the developed application met all functional requirements and provided a user-friendly experience for staff and administrators.

1.5 Scope and Limitations

Scope:

- Reservation Management – This feature delivered an interactive, functional module allowing both frontend lounge staff and registered clients to schedule and secure specific study spaces, study pods, or conference rooms in advance. It incorporated programmatic validation logic to prevent double-bookings, handle reservation overlaps, and optimize physical table allocation in real time.
- Membership and Time Tracking – This module securely recorded detailed customer profiles and automatically monitored physical usage durations via server-side check-in and check-out timestamps. It programmatically tracked subscription validity periods and membership tier benefits to ensure workspace privileges were validated automatically before a session began.
- Payment Billing – This component handled the internal financial workflow of the lounge. It dynamically converted the accumulated duration from the time logs into itemized financial charges based on the user's applicable hourly rate or membership status, ensuring all individual customer transactions were calculated and recorded accurately
- Sales Recording and Reporting – The system permanently logged all financial transactions to systematically compile analytical summaries. It generated real-time, tamper-proof sales reports categorized by daily, weekly, or monthly intervals, providing management with accurate ledger data to monitor business performance effectively.
- Admin and Staff Dashboards – The system featured distinct centralized interfaces tailored to specific user roles. The control screens displayed live metrics regarding room occupancy, active workspace

countdowns, real-time sales summaries, and automated time-expiration alerts to facilitate seamless front-desk operations and high-level administrative decision-making

- Role-Based Access Control (RBAC) – To reinforce system security and data integrity, strict access permissions were programmatically enforced based on user type. System administrators were granted full access to core configurations, analytical reports, and system parameters, while frontline staff members were restricted to purely operational features such as logging walk-ins, executing bookings, and processing billings

Limitations:

- Geographic Limitation – The system was exclusively designed and configured around the specific business model and localized workflows of the WS Students & Professionals Lounge – La Paz Branch. The operational rules, spatial layout, and membership structures were tailored to this single branch and did not support multi-branch scaling, cross-branch data synchronization, or shared roaming memberships.
- Platform Limitation – The application was strictly developed as a web-based system. Although the user interfaces were designed responsively to adapt to mobile screens (smartphones and tablets), the project scope completely excluded the development of native mobile applications for the Apple App Store (iOS) or Google Play Store (Android)
- Internet Dependency – Because the application relied on a cloud-hosted framework and a centralized relational database to prevent fragmented records, it exhibited absolute internet dependency. The system required an active, stable internet connection to process transactions and did not feature an offline-first synchronization mode or local peer-to-peer data backup capabilities during internet outages.
- No Accounting Integration – The software focused strictly on day-to-day operational logistics (reservations, memberships, time tracking, basic billing, and historical sales logs). It did not function as an enterprise resource planning (ERP) system and completely excluded complex accounting integrations, such as balance sheets, profit-and-loss statements, tax computations, or automated payroll deductions.
- No Hardware Automation – The platform operated purely at the software application layer. It did not integrate with physical Internet-of-Things (IoT) hardware, biometric fingerprint sensors, RFID card scanners at the entrance, or automated electronic door lock strikes for the study rooms. Physical access verification and final check-in logging remained under the manual oversight of front-desk personnel.

1.6 Definition of Terms

Administrator – One who manages computer and network systems on behalf of another, such as an employer or a client. Administrators are the people who make things work and keep it all running [1].

In this study, the term refers to the management or authorized personnel of WS Students & Professionals Lounge – La Paz Branch who possess the highest level of system configuration privileges [1]. Within the platform, this role is responsible for overseeing business operations, managing user access credentials, generating consolidated sales reports, and auditing historical system logs to ensure operational accountability [1].

Analytical Reports – A document that combines data and analysis to help an organization evaluate business decisions, identify trends, or measure operational performance [2].

In this study, the term refers to the systematically generated data summaries, visualizations, and tabular charts within the system that translated raw operational logs into actionable business insights [2]. These reports compiled and analyzed aggregated system metrics to visually map out lounge space usage trends, peak operating hours, and customer demographic metrics for administrative decision-making [2].

Centralized System – A computing architecture where all data processing, storage, and management control are concentrated within a single primary node or server network [3].

In this study, the term refers to the unified platform where all operations of the lounge—such as space booking, membership logs, and payment data—were integrated into a single database to eliminate fragmented data records [3].

Database – An organized, digitally stored collection of structured data managed through a software framework to allow rapid search and retrieval [4].

In this study, the term refers to the relational storage architecture developed using MySQL where all structured information critical to the lounge's operations was systematically stored and protected [4]. This included the secure preservation and real-time retrieval of interconnected tables concerning lounge members' profiles, physical check-in/check-out timestamps, historical financial transactions, and authorized staff accounts [4].

Membership Tracking – The systematic process and software automation used to monitor profile creation, validity status, and active engagements of registered clients within an organization [5].

In this study, the term refers to a dedicated functional module within the system that actively monitored and managed subscription lifecycles, user profiles, and associated tier benefits of long-term clients [5]. It automatically updated subscription validity windows, tracked recurring billing cycles, and validated active statuses during the check-in process [5].

Operational Bottlenecks – Any point of congestion in an enterprise workflow or service delivery line that reduces the speed and increases the error rate of the entire operation [6].

In this study, the term refers to the specific workflow delays, structural inefficiencies, and administrative errors experienced in the previous manual setup of the lounge, such as accidental double-bookings of study rooms, human mathematical errors during checkout time computations, and slow paper-based invoicing [6].

Reservation System – An automated software application that facilitates the booking, allocation, and scheduling of specific physical spaces or inventory items ahead of actual usage [7].

In this study, it refers to the interactive component of the web platform that permitted both frontline staff and general clients to securely book specific study pods or conference rooms in advance using real-time availability calendars and automated overlap validation logic [7].

Role-Based Access Control (RBAC) – A security mechanism that restricts system access and operations to authorized users based on their specific organizational role or assignment level [8].

In this study, it refers to the built-in security feature that restricted data access and modification privileges based on a defined user hierarchy: Administrator (full system configuration and reports), Front-desk Staff (check-ins, check-outs, and billing), and General Customers (viewing availability, reservations, and membership status) [8].

Sales Report – A structured accounting or transaction summary that reflects the revenue, volumes, and financial inputs of a firm over a designated timeline [9].

In this study, it refers to the auto-generated financial audit summary within the system that programmatically compiled and aggregated revenue earned over daily, weekly, or monthly intervals from workspace room rentals and monthly membership subscriptions [9].

Study Hub Management System - A specialized information system engineered to automate workflows, client handling, asset scheduling, and billing pipelines within shared-workspace facilities [10].

In this study, it refers to the specific web based software platform developed for the WS Students & Professionals Lounge (La Paz Branch) to replace manual administrative workflows by centralizing user registrations, space reservations, membership subscriptions, time-based logging, and billing [10].

Time Logs – A chronological, tamper-evident digital record that captures explicit timestamps marking the execution and conclusion of user operations or resource allocation [11].

In this study, it refers to the automated database records that programmatically captured and stored the exact check-in and check-out timestamps of clients to dynamically calculate precise workspace usage durations and convert them into itemized billing charges [11].

User Dashboard – A consolidated, interactive graphical user interface (GUI) designed to present real-time summary data and essential functional entry points customized to a user's role [12].

In this study, it refers to the personalized control screen accessible by authenticated lounge clients upon logging in, allowing them to independently view active room reservations, track remaining membership validity days, and browse past transaction histories [12].

Web-Based System – A software system running on a remote host architecture that can be rendered and utilized through a web browser using client-server requests via HTTP/HTTPS [13].

In this study, it refers to the architectural deployment framework of the application, which allowed the lounge management platform o be securely accessed over an internet connection using standard web browsers across laptops, tablets, and smartphones without local software installations [13].

1.7 Conceptual Framework

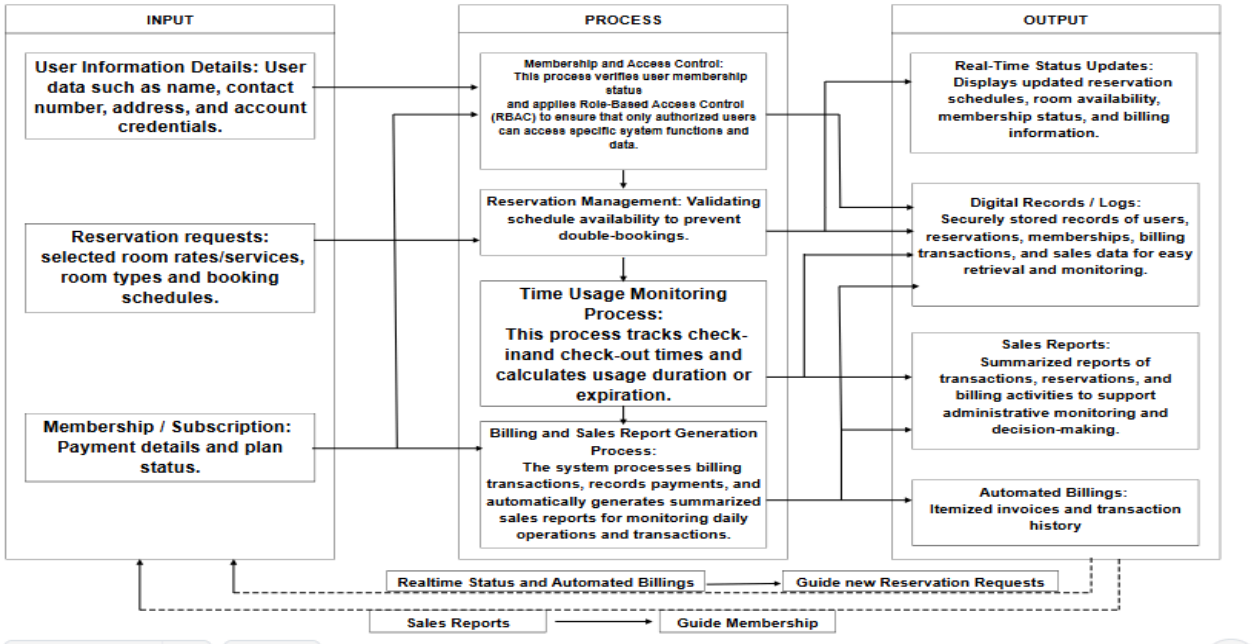


Figure 1. Conceptual Framework of the Integrated Study Hub Management System for WS Students & Professionals Lounge

The conceptual framework of the Integrated Study Hub Management System is anchored on the Input-Process-Output (IPO) model, illustrating the transformation of raw operational inputs into centralized, data-driven management outputs.

- INPUT: This phase captures the primary data requirements fed into the system:
    - User Information Details: Includes user identification, contact numbers, physical addresses, and account credentials required for profile setup and authentication.
    - Reservation Requests: Encompasses selected room rates, amenity services, room types, and preferred booking schedules.
    - Membership / Subscription: Consists of payment transaction details, tier selections, and current plan validity statuses.
- PROCESS: The process phase represents the computational logic and module workflows of the system executed in a sequential pipeline:
    - Membership and Access Control: Verifies user membership validity and applies Role-Based Access Control (RBAC) to ensure that system privileges are strictly restricted according to user roles.
    - Reservation Management: Programmatically validates schedule availability and time slot overlaps to prevent resource double-bookings.
    - Time Usage Monitoring Process: Tracks dynamic check-in and check-out timestamps to compute exact session usage durations or subscription expiration limits.
    - Billing and Sales Report Generation Process: Processes itemized financial charges based on duration and rates, logs payments, and compiles aggregated transaction data.
- OUTPUT: The system produces four main deliverables designed to streamline hub operations:
    - Real-Time Status Updates: Displays live schedules, current room availability, membership validity, and billing details.
    - Digital Records / Logs: Securely stores indexed relational data concerning users, bookings, subscriptions, transactions, and system audit logs for rapid retrieval.
    - Sales Reports: Generates summarized analytical reports of financial activities and occupancy trends to assist management in strategic decision-making.
    - Automated Billings: Issues itemized customer invoices and records transaction histories automatically.
- FEEDBACK LOOPS: To foster continuous operational efficiency, the framework incorporates two distinct feedback mechanisms:
    - *Feedback 1:* Outputs from Real-Time Status and Automated Billings cycle back to inform and refine new Reservation Requests by providing prospective clients with immediate availability and pricing clarity.
    - *Feedback 2:* Analytical insights from Sales Reports loop back to guide administrative decisions regarding Membership / Subscription tier adjustments, pricing updates, and promotional strategies.

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CHAPTER II

REVIEW RELATED LITERATURE

A study hub management system represents a specialized class of administrative software designed to orchestrate, automate, and centralize the diverse operational workflows within modern co-working and shared-facility workspaces [1]. In contemporary business and academic environments, these systems have evolved from simple scheduling applications into comprehensive platform ecosystems [1]. They allow business owners, frontline personnel, and clients to seamlessly coordinate resource allocations, handle memberships, and process financial exchanges regardless of location [1].

To cater to varying business demands, workspace management systems utilize diverse digital infrastructures:

- Cloud-Hosted Booking Infrastructures – A framework that provides rapid, remote access to real-time space availability, allowing clients to secure reservations dynamically via any internet-enabled device [2].

- Cross-Platform Mobile Integration – A method optimized for mobile devices, enabling clients to track their remaining membership duration, receive real-time notifications, and monitor current check-in statuses on the go [2].
- Administrative Desktop Command Centers – A stationary control platform engineered for frontline lounge staff, utilized to handle walk-in clients, audit facility check-outs, and monitor overall utility usage within the hub [2].
- Local Relational Databases – A structured data architecture that serves as the backbone of the lounge, securely holding financial logs, user authentication files, and active timetracking sheets to ensure high data consistency [3].

In the context of daily business operations within coworking facilities, relying on manual transaction frameworks creates significant operational bottlenecks. Paper-based logbooks and fragmented communication channels—such as manual social media message logging—are heavily susceptible to human oversight errors. These errors frequently manifest as overlapping space reservations (double bookings), inaccurate manual timekeeping, disorganized customer profile tracking, and slow invoice generation [4]. For example, when multiple clients attempt to secure a group conference room simultaneously without a real-time ledger, it often leads to operational conflicts, scheduling delays, and an eventual decline in customer trust [4].

The strategic deployment of an integrated, automated management system mitigates these administrative risks by enforcing precise digital validation rules. By automating the lifespan of a booking—from the initial online reservation to the final algorithmic billing calculations upon check-out—the system guarantees structural data integrity and transaction transparency []. Digital platforms drastically accelerate transaction speeds, lower the business's dependence on physically tracking paper logs, and eliminate risks concerning misplaced receipts or corrupted customer records [5]. By resolving these operational vulnerabilities, an integrated system protects enterprise financial data, enhances staff productivity, and guarantees a seamless, modern, and reliable environment for both student and professional clients [5].

2.1 Foreign Studies

Management Information Systems: Managing the Digital Firm. The study by Laudon and Laudon highlights management information systems (MIS) as critical tools that integrate people, processes, and technology to support organizational operations [6]. Applied to an integrated study hub, an MIS streamlines core workflow processes including reservations, attendance tracking, resource allocation, and user management [6]. Automating these routine operations minimizes manual effort and operational errors, thereby improving overall service quality and user satisfaction [6]. Furthermore, centralizing data collection enables administrators to analyze usage patterns and peak hours for optimized space utilization and strategic planning [6]. Ultimately, adopting digital solutions aligns with modern technological demands by delivering flexible, data-driven environments for working students and professionals [6].

Information Technology for Management Driving Digital Transformation to Increase Local and Global Performance, Growth and Sustainability Information Technology for Management is an excellent, comprehensive guide on how modern enterprises bridge the gap between core business strategy and emerging tech infrastructure [7]. The study of Turban, E., Pollard, C., & Wood, G. Digital Transformation Disrupts Companies, Competition, and Careers Locally and Globally [7]. Information Technology for Management emphasizes that modern enterprises must align business and IT strategies to support digital transformation in on-demand and sharing economies [7]. By replacing outdated legacy architectures with integrated digital technologies, mobile solutions, and smart devices, organizations can reduce maintenance costs, improve operational efficiency, and deliver immediate services [7]. To enhance performance and sustainability, IT initiatives focus on product development, stakeholder integration, process improvement, cost efficiencies, competitive advantage, and global connectivity [7]. Furthermore, leverage

of Business Process Re-engineering (BPR) enables organizations to radically redesign internal workflows rather than making minor incremental changes [7]. Ultimately, adopting these disruptive technologies allows firms to differentiate themselves, lower transaction costs, and provide higher-quality service [7].

Developing and Implementing Web-based Online University Facilities Reservation System The study by Alkhaldi et al. Community services have recently grown through government and private institutions due to an increased awareness of their importance in societal development[8]. These distinctive services are now available free of charge, expanding beyond financial assistance to encompass various other forms of community support[8]. Ultimately, these diverse services are vital for the progress and prosperity of society by helping individuals become active and constructive citizens[8]. Recognizing this responsibility, universities actively support public engagement by offering community initiatives and making campus facilities—such as event halls, auditoriums, swimming pools, and sports courts—accessible to staff, students, and local visitors [8]. However, managing venue access without an online, centralized platform leads to scheduling bottlenecks and imbalanced facility utilization [8]. To address this challenge, Alkhaldi et al. designed and implemented a centralized, web-based facility reservation system for Imam Abdulrahman Bin Faisal University (IAU) [8]. Developed using Unified Modeling Language (UML), MySQL, and Visual Basic (VB), the application provides a user-friendly interface that allows users to view real-time availability schedules, submit reservation requests, cancel bookings, and manage user accounts remotely [8]. By replacing manual processes with automated scheduling, the platform optimizes campus resource allocation and supports broader community outreach initiatives [8].

Restaurant Management System For Kas Valley Restaurant. The rapid advancement of technology has transformed business operations across global industries, including the food and hospitality sectors [9]. In contemporary economies, digital tools like web platforms and dedicated applications optimize operational workflows, maximize customer satisfaction, and protect businesses during unexpected market disruptions [9]. However, many establishments still struggle with manual or fragmented administrative processes [9]. Lacking a centralized digital architecture often leads to operational bottlenecks such as long physical queues, phone delays, inefficient table reservation tracking, and difficult sales data analysis [9].

To resolve these administrative vulnerabilities, Alormene developed a centralized, web-based restaurant management system tailored for Kas Valley Restaurant [9]. The platform replaces manual ordering and scheduling workflows by unifying online and in-person meal ordering, table reservation tracking, customer data management, and automated sales analytics within a single database system [9]. By eliminating data redundancy and fragmentations across operational units, the system streamlines daily business tasks, enhances customer satisfaction, and provides actionable data insights for executive decision-making [9].

Design and Implementation of Reservation Management System Case Study: Grand Ville Hotels. The advances in mobile technology and availability of broadband have made it easier to search for a hotel and book through an online reservation system [10]. Although mobile applications are unlikely to revolutionize a hotels' reservation system to the same extent that the internet did, it may have an impact on hotel revenue management and the concept of implementing price structures [10]. The management and booking of rooms in hospitality establishments becomes increasingly complex and prone to inefficiencies when reliant on manual tracking or fragmented tools such as standalone spreadsheets [10]. Managing customer records using disconnected Excel sheets often results in delayed information retrieval, slow data entry, difficult record updates, and time-consuming report compilation [10]. These administrative bottlenecks increase operational friction and make it challenging to maintain real-time availability tracking for prospective guests [10].

To overcome these operational limitations, Williams and Ajinaja developed a modernized hotel reservation system utilizing an object-oriented software development approach (OOAD)—incorporating object-oriented analysis, design, and programming to ensure system reliability, maintainability, and scalability [10]. The platform replaces fragmented logs by centralizing room reservations, customer profiles, and transaction records into a unified database

[10]. By integrating real-time booking capabilities with core management and accounting workflows, the system streamlines room allocations, optimizes revenue across sales channels, and empowers management to make data-driven decisions [10].

Design and Implementation of a Reservation System and a New Queuing for Remote Labs. According to Khazri, Y., Fahli, A., Moussetad, M., and Naddami, A. The fast development of the Internet technology and its increase of popularity has an enormous impact on the process of teaching [11]. Modern technology offers innovative tools that enhance educational strategies, with internet and mobile technologies fostering practical scientific and technical training through interactive applications [11]. E-learning and distributed computing systems have expanded practical education by offering remote laboratories as viable alternatives to traditional face-to-face practical work [11]. Environments like the ELAB FSBM platform complement conventional training architectures by enabling students to manipulate actual, complex, or hazardous experimental equipment from a distance [11].

This setup mitigates student massification, addresses shortages in physical laboratory space and coaching resources, and facilitates resource sharing across national and international institutions [11]. Furthermore, virtual and remote platforms lower operational costs by minimizing the need for physical equipment maintenance, expand accessibility for individuals with disabilities, and incorporate dynamic user queue management systems to coordinate collaborative, supervised experiments between instructors and learners [11].

Project Management in Prototyping an Online Table Reservation System. The rapid digitalization of the food and beverage industry has transformed operational management and customer interactions through smart, data-driven platforms for ordering, payment, and customer relations [12]. Despite widespread digitalization across service industries, many food and beverage (F&B) establishments continue to rely on manual workflows—such as phone calls, walk-ins, and unstructured messaging apps—to manage table reservations [12]. These manual frameworks routinely introduce operational friction, including miscommunication, double bookings, prolonged customer waiting times, and imprecise demand forecasting during peak business hours [12]. While existing literature explores web-based booking platforms and smart service tools, technological architecture and managerial implementation are frequently treated as isolated concerns [12]. Consequently, there is a distinct lack of empirical work demonstrating how standardized project management frameworks can systematically govern the technical development of online reservation platforms [12]. To bridge this domain gap, Felix et al. [12] developed a web-based table reservation system by integrating the Project Management Body of Knowledge (PMBOK) framework with the Systems Development Life Cycle (SDLC) [12]. By aligning core PMBOK knowledge areas—such as Scope, Time, Cost, Quality, and Risk Management—with concrete software development stages, the authors established a structured, dual-perspective reference model featuring Work Breakdown Structures (WBS), Gantt charts, and resource planning [12]. System evaluations confirmed that the resulting platform significantly reduced booking errors, shortened customer wait times, and improved table availability transparency, providing F&B managers with a practical, risk-mitigated blueprint for digital transformation [12].

Online Reservation Management System to Increase Transaction Efficiency at MOMENKITA Photo Studio. In an increasingly competitive business landscape, small and medium-sized enterprises (SMEs) in the creative sector require structured operational tools to maintain sustainable growth and operational efficiency [13]. Photo studio operations frequently encounter severe administrative friction when relying on manual management, resulting in double-booked photo sessions, unstructured customer profiles, untracked financial transactions, and reliance on physical print portfolios [13]. These manual constraints complicate session scheduling and hinder business expansion [13].To address these operational vulnerabilities, Firmansyah et al. [13] developed a web-based corporate management application tailored for "Momenkita Studio." The platform

unifies core studio workflows into a centralized digital system, incorporating an interactive availability calendar for online session bookings, secure digital payment processing, dynamic portfolio display galleries, and automated reporting modules for transactions and customer data [13]. Additionally, the application enforces role-based user access controls to streamline administrative oversight [13]. System evaluation demonstrated that visualizing schedule availability via an interactive calendar minimized booking overlaps, automated report generation, and significantly improved both customer experience and promotional reach [13].

Online Reservation Systems in E-Business: Analyzing Decision. Making in E-TourismThe study by Halkiopoulos et al. [14] focuses on the exploration of knowledge regarding online booking systems and the views of local student users concerning booking rates. The emergence of e-commerce has restructured traditional market dynamics by integrating telecommunications, digital payment architectures, automated inventory management, and strategic marketing into unified online platforms [14]. Within this ecosystem, e-tourism relies heavily on web-based booking tools to handle electronic data transfers, automate transaction processing, and facilitate direct, real-time interactions between service providers and end users [14]. While these platforms reduce administrative friction and optimize scheduling workflows, understanding how users navigate information searches, evaluate available options, and execute bookings remains essential for optimizing system usability and operational effectiveness [14].

Investigating user perspectives—particularly among digital-native demographics such as students—highlights the role of emotion-focused decision-making and cognitive factors during online reservation processes [14]. Evaluating customer characteristics alongside their digital touchpoints (from initial search to payment processing) demonstrates that technical infrastructure alone is insufficient; online management systems must align with user interaction patterns, provide transparent availability schedules, and streamline the transaction pipeline to ensure high satisfaction and operational success [14].

The Dentist Online Reservation System Design and Implementation Web Based Application and Database Management System Project Technology has changed many aspects of daily life, which has become increasingly dependent on network connectivity due to the expansion of the Internet. With online dentist reservation systems, processes become significantly faster and more efficient than traditional methods [15]. Thousands of businesses and service organizations have adopted online reservation systems to gain a competitive operational advantage over traditional, fragmented record-keeping [15]. Managing client appointments and scheduling workflows via disconnected spreadsheets, folders, and paper files often leads to delayed information access and high administrative overhead [15]. In contrast, web-based reservation architectures centralize database operations on secure remote servers, enabling administrators to rapidly access client records, execute real-time searches, and generate comprehensive administrative reports through standard web browsers [15].

To address these administrative challenges in healthcare service environments, Kerdvibulvech and Win [15] designed and implemented an online dentist reservation system utilizing ASP.NET, C#, SQL Server, and Microsoft Access. The system provides front-desk administrators with centralized tools to manage patient profiles, review appointment requests, view dynamic schedule calendars, and log patient feedback to continuously improve service delivery [15]. Furthermore, the authors designed the software using a modular architecture, establishing a flexible framework that allows future system expansions—such as automated online billing, dynamic calendar linking, and direct patient-side appointment rescheduling or cancellations—without destabilizing core system functions [15].

Web Site &Web Based Hotel Reservation Management System For Sky Lodge Hotel. Relaxation is often considered key to maintaining a high quality of life within modern lifestyles [16]. Rapid expansion in the tourism and hospitality sector requires service providers to adopt modern software solutions to manage growing operational demands [16]. Establishments operating via paper-based or manual record-keeping frameworks routinely experience

significant operational bottlenecks [16]. Increased administrative workload reduces task accuracy, leading to frequent room booking errors, imprecise billing calculations, slow check-in processing times, and potential data corruption or loss—factors that directly impair guest satisfaction and business revenue [16].

To address these administrative challenges, De Silva [16] designed and deployed a web-based hotel management system for The Sky Lodge Hotel in Gampola, Sri Lanka. The platform transitions the property's manual operations into a centralized, web-accessible database architecture [16]. Core modules streamline room availability tracking, online reservations, guest records, employee administration and leave tracking, operational expenditure logging, and system security controls [16]. By replacing manual paper logs with automated validation workflows, the platform eliminates scheduling overlaps, improves financial computation accuracy, safeguards customer data, and significantly enhances service efficiency for management, staff, and guests alike [16].

Development of a Web-Based Reservation Systems for Federal. Polytechnic Kaltungo Education is often regarded as the most suitable tool for national development, helping citizens build the skills and attitudes required for nation-building [17]. Gaps in practical skills among graduates present a significant challenge for higher education institutions, particularly in tourism and hospitality programs where employers require proficiency in digital reservation and booking operations [17]. In response to regulatory mandates—such as the National Board for Technical Education (NBTE) requirements in Nigeria—educational departments must provide students with hands-on exposure to functional reservation mechanisms to bridge the divide between theoretical instruction and real-world industry demands [17].

To address this training deficit, Abdulrauf et al. [17] developed a web-based reservation platform tailored for the Department of Tourism at Federal Polytechnic Kaltungo. Built using Unified Modeling Language (UML), PHP, HTML, CSS, and JavaScript, the application incorporates distinct functional modules allowing students to simulate online bookings for hotel rooms, airline tickets, and food orders within a single interface [17]. Evaluated by academic leadership, faculty members, and students through comprehensive testing, the system demonstrated strong usability and instructional effectiveness [17]. The findings confirm that integrating interactive, simulation-based web platforms into academic curricula significantly enhances students' practical skills, mitigates operational training bottlenecks, and aligns graduate competencies with modern digital service standards [17].

Understanding the Effects of Reservation Systems and Online. Transaction Capacity on the Competitiveness of Travel Agencies The rapid acceleration of globalization and information communication technologies (ICT) has established an information-driven economy where data serves as a critical strategic asset [18]. To navigate this evolving market, modern service organizations must strategically integrate digital infrastructure to ensure sustainable operations and meet expanding consumer expectations [18]. In the travel and hospitality sectors, relying exclusively on conventional office setups without automated digital capabilities creates competitive disadvantages, as consumers increasingly expect instant, web-accessible transactions and real-time information processing [18].

To evaluate these technological dynamics, Işkın et al. [18] investigated the impact of reservation systems and online transaction capacity on firm competitiveness using a quantitative survey of 566 travel agency managers in Istanbul. Their empirical findings confirmed that automated reservation systems and robust online transaction portals directly improve business competitiveness across three essential performance metrics: cost reduction, service quality enhancement, and market share expansion [18]. The study concludes that traditional travel agencies must adopt

integrated reservation architectures and digital processing tools to lower administrative costs, optimize service delivery, and maintain a sustainable competitive advantage in the modern market [18].

Development and Implementation of a Python-Based Hotel. Management System: A Comprehensive Tool for Room Reservation, Payment, and Administration The hospitality industry faces ongoing operational bottlenecks due to manual management processes, which frequently result in administrative delays, data entry errors, and security risks during room bookings, payment processing, and guest tracking [19]. To overcome these challenges, there is a strong demand for automated, centralized software architectures that streamline front-desk operations, ensure data integrity, and enhance overall service efficiency [19]. To address these operational vulnerabilities, Alam et al. designed and implemented a scalable, web-based hotel management system built on Python using the Django web framework and a MySQL relational database [19]. Developed by mapping clear functional and non-functional requirements, the platform features real-time reservation processing, secure financial transactions, and role-based access control (RBAC) to enforce isolated permissions for guests and system administrators [19]. Rigorous unit, integration, functional, and security testing confirmed that the system significantly reduces operational errors, accelerates daily administrative tasks, safeguards transaction data, and delivers an intuitive user experience for both management and guests [19].

2.2 Local Studies

Web-Based Reservation and Management System for Nature Spring Resort. Ramento, Dancel, Taberlo, and Anquillano. The significant improvements in accuracy, operational processes, and user contentment demonstrate the system's success [20].

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Online Reservation Inventory Systems in Higher Education: A Gap Analysis and Enhancement Framework with SMS Notification Support. Sayson, J,M., Reyes, J,C., and Pisueña, J,S. Inventory and reservation management systems are essential for maintaining seamless administrative and operational workflows within higher education institutions [21]. As a core component of resource management, physical asset tracking directly impacts institutional accountability, operational efficiency, and overall user satisfaction [21]. To optimize these outcomes, academic institutions increasingly adopt digital reservation and inventory platforms to track, allocate, and monitor shared assets—such as university equipment, specialized laboratories, and physical spaces [21]. Implementing centralized digital platforms ensures the structured data collection and documentation necessary for strict institutional compliance, operational oversight, and strategic resource planning [21].

While online inventory and booking systems are widely deployed across academic environments, their functional maturity and communication capabilities vary significantly [21]. Standard platforms effectively reduce manual record-keeping errors, prevent untracked inventory losses, and streamline resource allocation through core features such as item tracking, logs, automated reports, and role-based access control (RBAC) [21]. However, current implementations frequently suffer from notable operational gaps, including poor real-time user notification mechanisms, system accessibility limitations, and minimal integration with modern mobile communication tools [21].

To address these vulnerabilities, Sayson et al. [21] conducted a gap analysis and developed an enhancement framework using a Systematic Literature Review (SLR) and Design Science Research (DSR) approach. Their research specifically examined the operational impact of integrating real-time SMS notification modules into academic reservation architectures [21]. By delivering automated text alerts for booking confirmations, schedule updates, and status changes, the framework bridges communication gaps between administrators and users, minimizes missed reservations, and improves overall system responsiveness. The authors established a scalable, configurable blueprint designed to resolve recurring administrative bottlenecks in university resource allocation [21].

Digital Transformation of Reservation Services: Development of the Foxrock Internet Reservation Management System. Cajetas, M, M, Jr. et al. The rapid advancement of technology has fundamentally transformed the way organizations manage reservations and deliver customer services [22]. A growing number of businesses have adopted digital reservation platforms to streamline the booking process and reduce turnaround time. Such systems offer customers a significantly more convenient and accessible channel for scheduling appointments or securing reservations without the constraints of location or business hours, which in turn contributes to higher levels of customer satisfaction. Moreover, by automating booking transactions, these platforms substantially reduce the volume of incoming phone calls and email inquiries, allowing staff to redirect their attention toward higher-value operational tasks [22].

A Management Information System (MIS) serves as a unified platform that gathers, processes, stores, and distributes data to facilitate decision-making, organizational coordination, and operational oversight. The present study examines how MIS contributes to improving the efficiency of organizational decision-making by delivering timely data and in-depth analytical insights that align with institutional objectives while harnessing emerging technologies such as artificial intelligence and cloud infrastructure. By bringing together human resources, data assets, hardware, software systems, and established procedures, MIS enables organizations to streamline operations and empowers managers to arrive at well-informed strategic and day-to-day decisions [22].

The online reservation platform developed by Cajetas et al. [22] is expected to assist Foxrock Internet Services in modernizing and0 enhancing its existing service delivery mechanisms. Through the company’s user-friendly website, clients will be able to conveniently schedule internet installation appointments entirely online. Foxrock Internet Services, founded in 2018 by Mr. Kevin Paul Pacina and Mrs. Sharmaine Duraca Pacina, operates in Barangay Malbago, Madridejos, Cebu, and offers both fiber and wireless internet connectivity to its subscribers [22].

Numerous rural and underserved localities continue to suffer from poor or completely absent internet connectivity, commonly referred to as "dead zones." This lack of access severely curtails residents' capacity to engage with online services and leaves them uninformed about available connection options, underscoring the persistent digital divide separating rural communities from urban counterparts [22].

Prior research by Cajetas et al. [22] indicates that digital systems strengthen communication channels between service providers and clients while alleviating administrative burdens. By offering real-time availability information and automated booking confirmations, these platforms eliminate the need for repetitive follow-up inquiries. This automation significantly cuts the time and financial costs traditionally associated with manual scheduling workflows. In a standard system setup, customers simply supply their personal details, product preferences, and preferred payment option. Finally, settlement can be conveniently handled by field personnel or facilitated through flexible digital services like GCash to increase accessibility [22].

Organizations increasingly adopt digital reservation platforms to optimize their workflows and improve operational efficiency [22]. These modern systems reduce dependence on physical paperwork while minimizing scheduling conflicts like duplicate bookings. Additionally, they offer users the flexibility to submit reservations at any hour without being restricted by standard office hours. Despite these clear advantages, a number of organizations still rely on traditional manual booking methods that remain prone to errors. In contrast, the Foxrock Internet Reservation

Management System addresses these inefficiencies through clearly identified functional requirements and an intuitive, user-centered design [22].

The overall findings of the study reflect a highly favorable reception of the Foxrock Internet Reservation Management System across all evaluated dimensions [22]. Respondents assigned exceptional ratings ranging from "Very Satisfactory" to "Excellent" across a broad array of core functional criteria. These positive results confirm that the system successfully met, and in several respects surpassed, user expectations regarding its primary capabilities. The only identified area with notable potential for future growth was the responsiveness of the mobile application interface. Although the mobile experience still achieved a "Very Satisfactory" rating of 4.3, this specific score indicates a clear opportunity to further optimize performance and user engagement [22].

Development and Evaluation of a Web‐ Based Retail Management System for Small Business Operations. Dela Cruz and Santos [23] investigated the development of a web-based retail management system designed specifically for small businesses. In their study, they observed how shifting from manual transaction and inventory processes to an automated web-based system directly improved operational performance.

The system significantly enhanced transaction accuracy by minimizing errors commonly associated with manual data entry and recordkeeping [23]. Additionally, the system enabled real-time inventory tracking, helping business owners monitor stock levels instantly and make informed replenishment decisions [23]. Such automation reduced discrepancies between recorded and actual inventory, thereby enhancing overall business productivity and reliability [23].

These results are relevant to the current study because they demonstrate how automation and integration of digital systems can improve efficiency, accuracy, and data visibility—principles that support the need for an automated lounge management system with features like real-time scheduling, resource tracking, and user interactions [23].

SORMS: Steffi’s Online Reservation Management System. Balintag et al. [24] developed Steffi’s Online Reservation Management System (SORMS), a web-based digital solution designed specifically to transition Steffi’s Garden Resort away from the inefficiencies of a manual booking system. Manual reservation processes remain highly prone to errors regarding usability, functionality, safety, privacy, and information accuracy, frequently causing inconvenience for customers [24].

These inefficiencies highlight the necessity for service providers, such as hotels and resorts, to implement online reservation management systems [24]. Opting for direct software licensing is often more cost-effective for hoteliers

than paying recurring commissions to external third-party booking portals [24]. Furthermore, customers find it significantly more convenient to execute bookings at their leisure using automated digital platforms rather than relying on manual data entry [24].

Manual reservation procedures across various service sectors have consistently proven inefficient; in restaurant environments, for instance, customers frequently experience extended wait times for table availability that software-based platforms readily eliminate [24]. Establishing this digital infrastructure—comprising servers, network architecture, and software applications—aligns directly with Sustainable Development Goal (SDG) 9 by fostering resilient infrastructure, sustainable industrialization, and innovation [24]. Through digitalization, enterprises optimize daily operations and liberate technical resources for further innovation [24].

To stay competitive and improve service quality, businesses must continuously adapt to technological advancements [24]. Although reservation handling appears straightforward, manual execution involves operational bottlenecks that directly impair guest satisfaction [24]. SORMS addresses these vulnerabilities by providing a user-friendly platform that allows clients to seamlessly book and manage reservations online, reducing transaction errors and improving overall guest satisfaction [24].

The system architecture was constructed using the Agile Scrum methodology, with development structured around product backlog refinement, sprint planning, and iterative development cycles [24]. Architectural design was mapped using Gantt charts, Data Flow Diagrams (DFD), and Entity-Relationship Diagrams (ERD), supported by continuous feedback loops with resort staff and end-users to ensure practical relevance [24].

Managing Customer Reservations of BulSU Hostel through the Development of Online Information and Reservation System. his study by Castro et al. [25] offers practical insights into how technology can streamline service operations and improve customer experiences in institutional facilities. The authors explain that an online information and reservation system acts as a centralized platform that connects customers, staff, and facility resources to automate booking processes, ensure real-time availability tracking, and simplify administrative tasks [25].

This concept is highly relevant to the development of various facility management systems—such as library room reservation or hostel booking platforms—as such systems require effective integration of user access, resource monitoring, and service coordination to reduce errors, save time, and enhance overall operational efficiency [25]. Accurate and timely reservations are necessary to ensure proper service and positive customer experience for many service-oriented businesses [25].

In this study, the proponents considered factors like customer satisfaction and positive experience during a hostel stay, which begins with the reservation procedure [25]. To manage customer reservations for the Bulacan State University (BulSU) Hostel, the proponents designed and developed the "Online Information and Reservation System for BulSU Hostel" [25]. Different functionalities were incorporated into the system, including user account creation for staff and online customers, price updates through an online admin portal, bank payment processing, email confirmations, report generation (income, customer counts, and monthly summaries), page view counters, and configurable rate settings [25]. Visual Basic .NET and ASP.NET were used for the front-end, Microsoft Access served as the database application, and Crystal Reports was utilized for generating administrative reports [25].

The developed system was evaluated using the ISO 9126 software quality standard across criteria such as functionality, accuracy, reliability, user-friendliness, and security, with all metrics interpreted as "Acceptable" on a Likert scale [9]. Reserving a room guarantees guest availability upon arrival, fulfilling an institutional commitment [25]. Prior to internet services, travelers relied on written letters, direct phone calls, or travel agents; in contrast,

online reservations facilitate real-time arrangements and dynamic rate adjustments [25]. Founded in 2005, the BulSU Hostel provides budget-oriented accommodations for short-term course participants, conference attendees, foreign guests, and university students, while also serving as an on-the-job training facility for Hotel and Restaurant Administration students [25].

The study highlights that usability and user experience are central to the success of web-based reservation platforms [25]. The authors emphasize that the system features a clear interface and straightforward navigation, allowing personnel and guests to perform tasks easily regardless of technical background [25].

Web-Based Hotel Reservation System. The rapid expansion of global tourism has significantly raised operational standards across hospitality establishments, requiring even small properties to adopt structured administrative mechanisms [26]. Modern travelers depend heavily on well-organized, accessible reservation frameworks that facilitate efficient trip planning for both business and leisure contexts [26]. To address these operational needs, Olaño et al. [26] evaluated the reservation and billing workflows of Queens Hotel to identify opportunities for technological optimization.

Implementing a web-based hotel reservation system enhances operational reliability and accelerates administrative processes, particularly regarding customer room registration and automated cost estimation [26]. By automating core reservation workflows, digital platforms eliminate the complexity associated with manual booking and billing procedures [26]. Although transitioning to automated reservation software may require operational adjustments for administrative staff, automated calculation and booking management significantly reduce calculation errors and processing delays compared to manual methods [26].

A central reservation system (CRS) serves as a primary operational tool in hotel administration, providing a centralized framework for managing property availability, booking transactions, and sales channels [26]. By consolidating inventory and pricing data into a unified platform, hoteliers can maintain consistent availability tracking across various distribution channels, including direct online bookings and travel agencies [26].

Booking and Reservation System: A Unified Application Using Location-Based Services for Sustainable Tourism Networks. Gosela and Encarnacion [27]. developed the LakByahe Application, a unified web-based booking and reservation system designed to address tourism-related challenges in Nasipit, Agusan del Norte. The study identified the reliance on social media as a source of travel information and the lack of a centralized platform for local tourism information as significant gaps. To address these issues, the application integrates location-based services, streamlined booking and reservation processes, real-time tracking, transit solutions, and a directory of tourism establishments [27].

The researchers found that the proposed system can improve access to localized tourism information and facilitate interactions between tourists and local businesses. The application also provides stakeholders with an interactive dashboard and statistical reports to support the monitoring of tourism activities. Overall, the study demonstrates how an integrated digital platform can improve the efficiency and accessibility of tourism-related booking and information services while supporting local tourism development [27].

Effects of Computer Reservation System in the Operations of Travel Agencies. In the travel and tourism industry, computerized reservation tools—primarily known as Global Distribution Systems (GDS) or Central Reservation Systems (CRS)—serve as the core operational infrastructure [28]. Mastery of GDS platforms, such as Amadeus and Sabre, represents an essential technical competency for industry personnel across travel agencies, airlines, and tour operations [28]. Functionally, these architectures manage real-time service data, eliminate geographical barriers between service providers and intermediaries, and facilitate automated distribution logistics

across global travel channels [28]. While the continuous expansion of web connectivity has driven disintermediation and forced legacy providers to adapt to digital frameworks, technology implementation requires balancing automated efficiency with operational limitations [28].

To evaluate these technological dynamics, Felicen and Ylagan [28] assessed the implementation and impact of Computer Reservation Systems among travel agencies operating in Batangas City and Lipa City, Philippines. The investigation evaluated agency operational profiles, identified active reservation platforms, examined offered services, and analyzed the technical, human, and financial effects of CRS adoption [28].

The empirical findings revealed that a majority of the evaluated agencies operated for 6 to 10 years, maintained small operational teams of 3 to 4 employees, and relied predominantly on the Abacus CRS platform for flight ticketing, itinerary bookings, and tour packaging [28]. The system significantly enhanced agency efficiency and productivity by securing real-time airline inventory, streamlining transaction workflows, and connecting clients to multi-modal travel options [28]. Ultimately, the study demonstrated that incorporating hands-on CRS training into academic curricula and pre-internship orientation programs aligns student skill sets with industry operational standards, mitigating practical skill deficits in hospitality and tourism education [28].

2.3 Related Literature

Designing a Web-Based Restaurant Reservation Information System with Requirement Prototyping Method. Sutjiadi, R., Rahmawati, T., Kristianto, A., and Kanessa, F, T. The development of information technology is proliferating, along with the increasing human need for fast, precise, and accurate information. The role of information technology in supporting the business world will make it easier for business people to run their business [29].

Internet-based digital communication tools are currently increasing. This cannot be separated from several supporting factors, including the increasingly affordable price of digital communication devices, such as smartphones, tablets and laptops. In addition, internet users are currently also experiencing a relatively rapid increase.

The widespread use of internet tools is supported by affordable devices like smartphones, tablets, and laptops, alongside a growing online user base [29].Using technology helps companies change their business strategies, expand their market reach, and increase customer satisfaction [29]. In service businesses like restaurants, relying on manual records for customer orders, table reservations, and payments creates risks like lost data, human error, and slow service [29]. Moving to a centralized online database allows for faster data retrieval, automated error checking, and quick report generation to improve overall productivity [29].

To solve these problems, Sutjiadi et al. [29] built a web-based system for Wisata Kampung Kemiri that combines table reservations and food ordering into one platform. Using the Requirement Prototyping development method, the creators worked closely with users during the design process to make sure the system met their exact needs [29]. Black-Box testing and User Acceptance Testing (UAT) confirmed that the reservation and ordering features worked smoothly [29]. The study shows that prototyping helps build user-friendly software and creates a strong foundation for future upgrades like mobile apps and online payments [29].

Operational Efficiency in Retail: Using Data Analytics to Optimize Inventory and Supply Chain Management. Data analytics plays a key role in retail by helping businesses balance customer demand with lower operational costs [30]. By analyzing sales history and market trends, retailers can improve demand forecasting to keep stock at ideal levels. Using methods like ABC analysis, safety stock calculations, and Economic Order Quantity (EOQ) helps prevent both overstocking and running out of items [30]. Additionally, tracking supplier performance and using location data for shipment tracking optimizes supply chain logistics [30].

Famoti et al. [30] highlight that using modern inventory software and warehouse management systems improves daily operations. These tools give businesses access to data visualization, reporting features, and predictive analytics for better decision-making [30]. Implementing these platforms helps companies replace guesswork with real data, leading to faster shipping, improved customer loyalty, and smarter business choices [30].

In conclusion, adopting data-driven strategies allows retail businesses to lower costs and stay competitive in a changing market [30]. Investing in proper inventory infrastructure and analytics tools provides a strong foundation for long-term growth and operational success [30].

System Development Life Cycle (SDLC). Structured software methodologies, such as the System Development Life Cycle (SDLC), provide a step-by-step framework for system development [31]. Following the SDLC improves overall software quality, lowers project risks, and ensures that all user requirements are properly met [31].

This framework directly supports the development of the study hub system by providing a clear structure for its design and implementation [31]. By following the phases of the SDLC, key features—such as real-time scheduling, billing, and reporting modules—can be systematically built, thoroughly tested, and validated before final deployment [31].

IOT Environmental-monitoring System Development. Mosquitoes pose a major threat to humans and animals by transmitting dangerous viruses and parasites such as West Nile, dengue, Zika, and malaria [32]. Because broad macro-level weather data often fails to provide accurate information for specific, localized areas, monitoring precise microclimate conditions is necessary to gather the high-resolution data needed for effective population control [32].

To achieve high-resolution data collection, Hur [32] led a multidisciplinary initiative at Texas A&M University to develop a custom Internet of Things (IoT) microclimate monitoring system. The project integrated research and engineering education by forming three undergraduate capstone teams—Big Dog Engineering, AIM-N, and Pond Hoppers—totaling 12 students supported by graduate mentors [32].

The three student teams designed complementary subsystems for the project [32]:

Engineering developed an autonomous mobile weather station to collect localized climate data.

AIM-N built a custom IoT network and a low-cost Raspberry Pi cluster server for sensor data processing and machine learning analysis. Pond Hoppers constructed a specialized quadcopter drone capable of landing on

standing water to collect site samples. The project demonstrated that integrating distinct, student-built IoT devices could successfully generate mobile, high-resolution environmental data to support advanced mosquito control research [32].

Reservation System and Customer Management Optimization. The entertainment industry, particularly recreational karaoke services, requires efficient operational tools to manage room bookings [33]. In many venues, room reservations are still handled manually using paper logs or simple forms, which frequently leads to inaccurate records, schedule conflicts, calculation errors, and lost transaction histories [33]. These manual limitations reduce operational staff efficiency and negatively affect the overall customer experience [33].

To address these operational challenges, Siregar et al. [33] investigated the development and implementation of a computerized reservation system for karaoke venues. Transitioning to a structured digital system provides real-time visibility into room availability, speeds up booking transactions, and minimizes data entry errors, such as duplicate bookings or incorrect customer details [1]. Full integration between customer profiles, room allocations, and payment tracking streamlines daily workflows and improves service reliability [33].

The study demonstrates that replacing manual booking methods with a computerized reservation framework significantly enhances operational accuracy, transparency, and data consistency [33]. The authors highlight that this digital architecture provides a practical model for service-based reservation management, establishing a solid foundation for future enhancements such as online booking capabilities, digital payment gateways, and data analytics modules [33].

Web Development Technologies. Documentation from MySQL and W3Schools, modern web applications rely on a combination of database management systems (MySQL) and server side scripting (PHP). MySQL is used for storing and retrieving structured data efficiently, while PHP enables dynamic content generation and secure transaction handling [34].

This information is essential to the proponents as it justifies the choice of the technology stack to be used. Using these tools allows developers to build a system that is scalable, secure, and accessible across devices[34].

Development Student-Teacher Online Booking Appointment System in Academic Institutions. Traditional manual appointment scheduling in academic institutions is often time-consuming, disorganized, and prone to schedule conflicts [35]. To resolve these operational challenges, Bello et al. [35] developed a web-based student-teacher appointment booking system built using PHP, JavaScript, and MySQL. The platform enables students to view real-time teacher availability, submit appointment requests with specific visit notes, and modify or cancel bookings [35]. Teachers can manage their availability schedules, review incoming requests, and send direct updates to students [35].

The system uses a two-tier client-server architecture communicating over HTTP, where the web server handles user authentication, dynamic page rendering, and database transactions [35]. Centralizing these administrative tasks within a user-friendly digital interface eliminates physical waiting times, increases daily student coverage, and provides a secure, retrievable record of all scheduled appointments [35].

Development of Online Student Course Registration System. Traditional student course registration is a labor-intensive and time-consuming process that relies heavily on physical paperwork and manual signatures from multiple academic officials [36]. To address these administrative bottlenecks, Singh et al. [36] developed an automated, web-based course registration system using PHP, jQuery, Apache, and MySQL.

The system is structured into five core modules—Admin, Masters, Transactions, Reports, and Utilities—and features secure, role-based access for students, faculty, and administrators [36]. By centralizing records in a digital database, the platform eliminates the need for physical presence, minimizes data duplication, and allows students to seamlessly register for courses remotely [36].

Implementing this digital architecture significantly reduces the administrative workload for both students and institutional staff while improving data accuracy and transparency [15]. The study concludes that transitioning to automated management systems is essential for modern educational institutions to optimize academic workflows, generate accurate reports, and improve overall service delivery [36].

An Analysis of Restaurant Table Reservations Systems. The increasing demand for digital services has made automated table reservation systems essential for modern restaurant operations [37]. Traditional seating management methods often struggle to balance customer preferences with table capacity, leading to extended wait times and lower customer satisfaction [37]. To address these challenges, digital reservation architectures streamline booking workflows, optimize table utilization, and improve service delivery across the dining industry [37].

Mahajan et al. [37] designed and implemented a python-based restaurant table reservation platform to modernize seating management. The system features a user-friendly interface that allows customers to view real-time table availability, select preferred dining times, specify seating preferences, and receive automated booking confirmations [37]. Backend database management handles reservation records securely, allowing restaurant personnel to manage multiple bookings, process cancellations, and track seating capacity in real time [37].

Implementing automated reservation systems provides long-term operational and financial benefits for hospitality businesses [37]. By automating seating workflows, restaurants reduce guest waiting times, improve resource allocation, and optimize table turnover rates to increase overall profitability [37]. The study concludes that maintaining continuous updates and regular system monitoring ensures digital platforms adapt to changing industry demands and evolving customer expectations [37].

Impact of Business Intelligence and Analytics on Decision. Making in Online Reservation Systems within the Hospitality Sector. The rapid growth of the global hospitality sector requires businesses to continuously innovate and deliver high-quality services to remain competitive [38]. Technological advancements—such as online reservation systems, mobile applications, and digital guest engagement tools—have transformed daily operations and empowered consumers to share real-time feedback [38]. To manage these expanding operations, organizations increasingly rely on data-driven decision-making to gain a competitive advantage [38].

Ukandu et al. [38] investigated the impact of Business Intelligence and Analytics (BIA) on online reservation platforms within the hospitality sector. By synthesizing research across multiple BIA implementations, the authors evaluated how extracting insights from operational data directly influences strategic decision-making

and business performance [38]. Integrating BIA tools enables reservation platforms to deliver personalized customer experiences and recommendation features while resolving operational bottlenecks in real time [38].

Implementing business intelligence and analytics allows organizations to unlock the full potential of their online reservation systems [38]. Consolidating data from multiple operational channels provides actionable insights that improve decision-making, optimize resource allocation, and drive overall revenue growth [38].

Reservation Systems As A Tool of Tourist Services Marketing. Modern reservation systems play a foundational role in the tourism and hospitality industry by simplifying how customers select, book, and manage travel services [39]. In a highly competitive market, tourism enterprises rely on multifunctional reservation platforms to access real-time market data, track customer preferences, and expand their distribution networks [39]. These digital platforms enhance business competitiveness by streamlining booking processes, offering secure payment options, and enabling active service promotion [39].

Myskiv and Wojtan [39] analyzed the global travel reservation market to define the operational hierarchy and distribution channels of modern booking systems. The authors categorized reservation architectures into five distinct operational levels based on geographic reach and system complexity [39]:

- First (Global) Level: Global Distribution Systems (GDS) managing international travel inventory.
- Second (Regional) Level: Online Travel Agencies (OTA) and Restaurant Reservation Systems (RRS).
- Third (Corporate) Level: Corporate Central Reservation Systems (CRS).
- Fourth (Local Automated) Level: Local automated booking websites.
- Fifth (Local Non-Automated) Level: Local call centers and manual intake channels.

The study demonstrates that computerized reservation systems serve as vital marketing and operational tools [39]. By providing multi-channel booking options across flights, lodging, and local transit, these digital networks allow travel enterprises to increase operational efficiency, broaden market reach, and deliver high-quality hospitality services [39].

Hotel Reservation Management System. The rapid growth of the tourism industry has driven hospitality enterprises to adopt Information and Communication Technologies (ICT) to streamline operations and maintain business competitiveness [40]. Many small and mid-sized hotels rely heavily on third-party Online Travel Agencies (OTAs) or basic informational websites, which reduces profit margins through high commission fees and limits direct guest engagement [40].

To address these operational limitations, Shun and Hassim [40] developed a web-based hotel reservation management system for Hotel Time Johor Bahru using the Waterfall model and a structured development approach. The system integrates five domain modules—management, reporting, inventory, reservation, and user registration—tailored across five user roles, including administrators, staff tiers, and guests [40]. The platform automates core operations such as room and hall bookings, price updates, transaction processing, inventory tracking, and financial report generation [40].

Implementing a direct online reservation platform enables hotel owners to avoid continuous OTA commission fees, thereby increasing net profitability and brand credibility [40]. The platform allows guests to compare rates and make secure online reservations while providing staff with centralized administrative tools [40]. However, the study notes that without a channel manager to synchronize inventory with external booking

networks in real time, risks of overbooking remain—highlighting the need for future integrations such as automated channel management, mobile application support, and expanded booking features [40]..

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study utilized a developmental research design because the primary goal of the proponents was to design, engineer, and evaluate a functional software system that resolved the operational bottlenecks of the WS Students & Professionals Lounge – La Paz Branch. The proponents focused on creating a practical, reliable, and deployable digital tool tailored for the actual operational environment of the enterprise.

A descriptive research approach was also employed to systematically analyze the pre-existing operational state of the lounge. This involved investigating how frontline personnel previously managed daily transactions, identifying recurring operational bottlenecks, and determining the specific functional requirements necessary for the system. By thoroughly documenting and describing the legacy manual processes, the proponents successfully designed a targeted digital solution that aligned with the operational needs of the lounge administrator, staff, and clients.

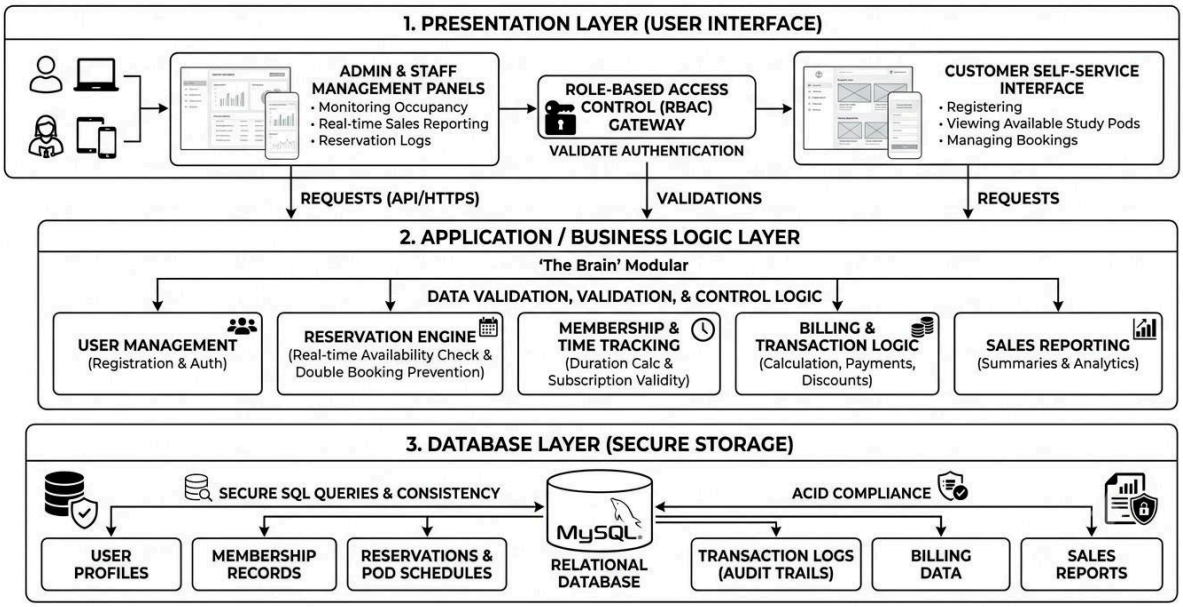
3.2 System Architecture

The Integrated Study Hub Management System was developed using a Three-Tier Architecture to ensure a structured, scalable, and organized flow of data across all operations. This architecture consisted of the following three core layers:

- Presentation Layer – Provided the Graphical User Interface (GUI) through which administrators, frontline staff, and customers interacted with the system. It featured a responsive web 26 dashboard for user account registration, real-time room occupancy monitoring, space reservations, and membership management. For administrative personnel and frontline staff, it included dedicated control panels for managing user profiles, auditing reservations, updating subscription tiers, processing check-in/check-out logs, and generating sales reports. This layer enforced Role-Based Access Control (RBAC) to dynamically customize navigation menus and functional views based on authenticated user roles—restricting sensitive configuration functions to authorized personnel while delivering an intuitive, self-service portal for general clients.
- Application / Logic Layer – Served as the core processing engine of the platform, handling all client-server requests and enforcing business logic rules. Key operational modules in this layer included:
- User Management Module – Handled account registration, credential encryption, and session authentication.
- Reservation Module – Checked real-time workspace availability and programmatically prevented schedule overlaps or double-bookings.
- Membership Module – Managed subscription lifecycles, calculated active session durations, and validated tier discounts. • Billing Module – Executed automated financial calculations based on time logs, hourly rates, and membership privileges.
- Sales Reporting Module – Aggregated transactional logs to generate itemized financial summaries.

This layer ensured that all user actions were validated prior to querying the database layer and returning processed results to the presentation interface.

- Database Layer – Managed the secure, relational storage of all core system records using MySQL. It housed interconnected tables storing user authentication profiles, membership subscription states, room availability schedules, time-in/time-out logs, billing transaction histories, and sales reports. The database layer maintained strict data integrity, consistency, and security, ensuring that every operational transaction was captured accurately and enabling rapid data retrieval for real-time reporting, auditing, and administrative decision-making.



3.3 System Development Methodology

The project followed the System Development Life Cycle (SDLC) methodology, executing the following sequential phases:

- Planning – In this phase, the proponents identified the core operational vulnerabilities in the manual workflow of the WS Students & Professionals Lounge – La Paz Branch, such as reservation conflicts and the difficulties of tracking active memberships, calculating time-based billing, and compiling daily sales reports. The proponents defined the system requirements and operational scope to ensure the project directly resolved manual recording errors, data fragmentation, and slow transaction processing speeds.
- Analysis – The proponents gathered operational data through structured surveys, stakeholder interviews, and direct field observations involving the administrators, frontline staff, and clients of the lounge. This involved analyzing the systemic risks associated with paper-based logbooks, such as physical data loss, room double-bookings, and the labor-intensive process of manually auditing usage durations and financial transactions.
- Design – The proponents engineered an intuitive, user-friendly also involved designing a normalized relational database schema (MySQL) capable of securely storing customer profiles, membership subscription states, reservation records, hourly rate matrices, billing logs, and sales transaction data to enable fast search and retrieval.

- Implementation – The proponents developed the software application and configured the backend database infrastructure. Key operational modules were implemented, including user registration, real-time reservation management, automated membership tracking, time-based room usage session monitoring, dynamic billing computation, and sales report generation. System security controls—specifically Role-Based Access Control (RBAC) and user authentication logs—were integrated to protect confidential enterprise data.
- Testing – The proponents executed comprehensive software testing across all modules to identify and rectify bugs, logical inconsistencies, or edge cases. Functionality was validated for user registration, space booking, membership verification, billing calculations, and reporting modules. Additionally, frontline personnel and administrators performed hands-on testing to evaluate system usability and confirm alignment with operational workflows.
- Deployment – In this phase, the system was deployed for actual operational use within the WS Students & Professionals Lounge – La Paz Branch. The proponents conducted user training sessions for frontline employees to ensure confident, proficient, and secure system operation. Post-deployment monitoring was conducted to verify operational stability and support personnel during initial business adoption.

3.4 Agile Development Methodology

The Agile Methodology was selected for the development of the Integrated Study Hub Management System because the project required a secured, organized, and highly adaptive framework tailored for the WS Students & Professionals Lounge – La Paz Branch. Agile provided the necessary flexibility to adapt to evolving project requirements, which proved essential as functional rules concerning user registration, interactive reservations, automated membership tracking, time-based usage session logging, billing computations, and sales reporting were continuously refined during development. This iterative approach enabled the proponents to receive continuous feedback from lounge management, ensuring that the final platform directly



Figure 2. Agile Development Methodology fostered close collaboration between the proponents and stakeholders, allowing technical challenges and edge cases to be identified and resolved early in the development lifecycle. In the Requirements Analysis phase, the proponents gathered detailed operational requirements regarding the lounge's legacy manual setup. This involved identifying specific administrative pain points, such as room double-bookings, manual check-in transcription errors, and difficulties in monitoring active membership validity windows.

The proponents formally defined the software requirements and system scope to ensure that the developed platform directly resolved the daily challenges faced by frontline staff and administrators. During the System Design phase, the proponents engineered the overall software architecture, module workflows, and database structures. This involved designing intuitive graphical user interfaces (GUIs) for the reservation, membership management, dynamic time-tracking, and billing modules. Additionally, a normalized relational database schema (MySQL) was constructed to securely store user credentials, membership statuses, booking calendars, time logs, itemized billing records, and sales summaries. The UI/UX design focused on simplicity and ease of use to minimize the learning curve for staff during the transition from paper-based operations to a digital platform.

In the Development phase, which encompassed active programming and database integration, the proponents constructed 25 the system according to the finalized architecture. Core functional modules—including user registration, real-time reservation validation, automated membership tracking, time-duration session logging, dynamic billing computation, and sales report generation—were programmed and integrated.

The proponents ensured that the underlying backend architecture was stable, secure, and capable of processing concurrent transactions without data corruption or latency. During the Testing phase, the proponents conducted rigorous system evaluation to verify functional accuracy and eliminate software defects. Testing validated the precision of automated billing calculations, the reliability of time-based session tracking, and the effectiveness of programmatic validation logic in preventing reservation overlaps.

Furthermore, the administrator and frontline personnel of WS Students & Professionals Lounge participated in formal User Acceptance Testing (UAT) to evaluate system usability, verify workflow alignment, and recommend minor interface refinements prior to deployment. Finally, in the Deployment phase, the proponents delivered the fully functional system to the WS Students & Professionals Lounge.

The proponents conducted structured orientation and hands-on training sessions for frontline employees to ensure confident and secure system operation. Initial post-deployment monitoring was maintained to verify system stability, resolve minor operational queries, and confirm that the platform was fully prepared to handle the lounge's daily operational transactions.

3.5 Data Collection Techniques

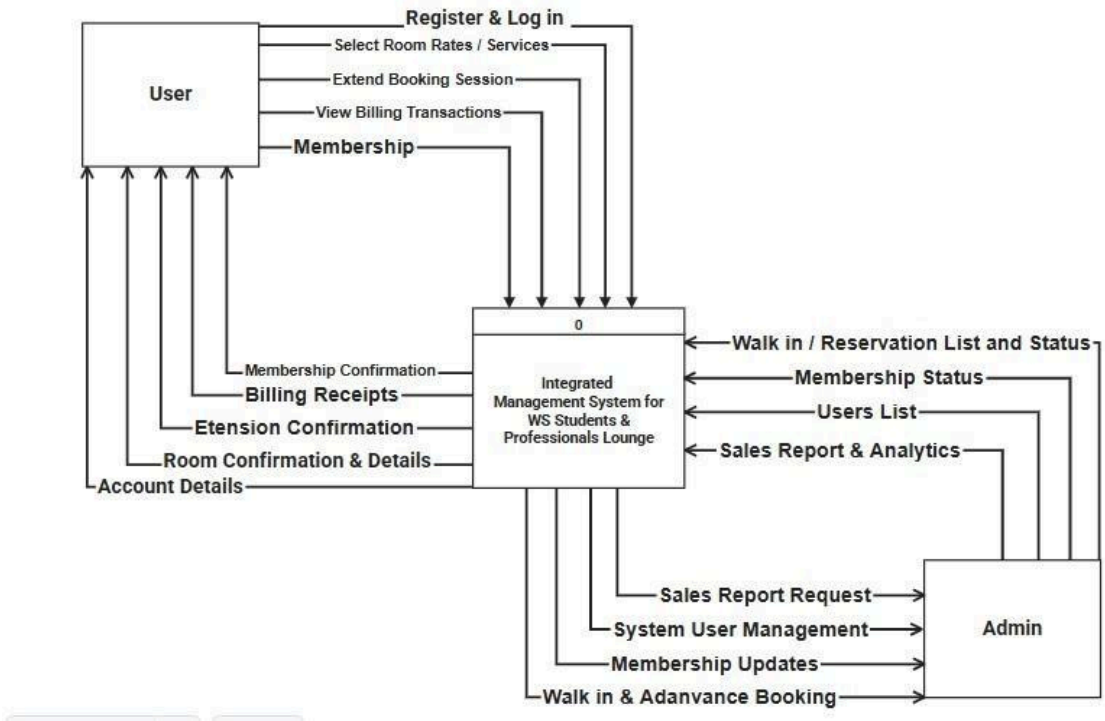
To design and develop an Integrated Study Hub Management System that addressed the specific operational needs of the management, frontline personnel, and clients of WS Students & Professionals Lounge – La Paz Branch, the proponents gathered primary data and operational insights using a combination of qualitative and quantitative collection methodologies. These techniques ensured that the developed platform was user-friendly, structurally sound, and directly aligned with the operational challenges identified during preliminary investigation.

1. Surveys The proponents distributed structured survey questionnaires to the target user groups, including the lounge owner, general manager, frontline staff, and registered clients. These surveys evaluated how respondents previously handled space reservations, membership tracking, billing transactions, and sales reporting. The instruments also assessed user perceptions regarding operational friction, task efficiency, and preferred software capabilities, providing quantitative data that guided subsequent system development.
2. Interviews The proponents conducted one-on-one structured and semi-structured interviews with the lounge administrator and key staff members involved in managing daily front-desk operations, room scheduling, billing, and record-keeping. This provided detailed qualitative insights into specific operational bottlenecks—such as manual calculation errors and scheduling overlaps—that quantitative surveys alone could not capture. Interviewees shared firsthand experiences, operational concerns, and recommendations for improving workflow efficiency.
3. Focus Group Discussions The proponents conducted focus group discussions (FGDs) with the lounge staff and management to review proposed system features and workflow designs. These collaborative sessions allowed participants to express opinions, evaluate prototype mockups, and highlight critical operational requirements. The feedback gathered during these discussions enabled the proponents to prioritize essential systemfeatures and refine UI/UX elements based on direct user consensus.
4. Observations To gain an accurate understanding of daily operational routines, the proponents performed direct field observations of the legacy manual processes at the lounge. The proponents monitored how staff recorded member profiles, verified study pod availability, calculated active customer usage session durations, processed manual billing transactions, and compiled end-of-day sales totals. Direct observation enabled the proponents to pinpoint precise stages in the manual workflow where delays, mathematical mistakes, and record duplication frequently occurred.
5. Document Analysis The proponents analyzed existing physical logbooks, manual reservation ledgers, billing receipts, sales summary sheets, and technical literature related to study hub operations and Point of

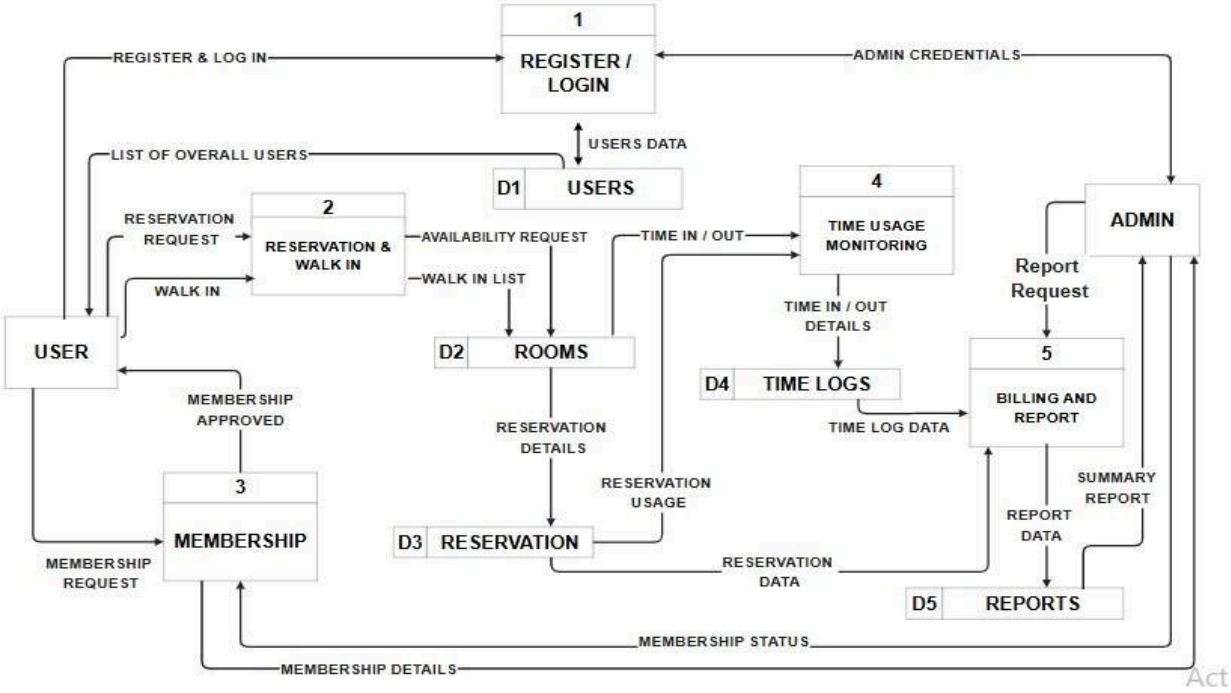
Sale (POS) architecture. Examining these historical records provided the proponents with a clear baseline of the lounge's data tracking structures, facilitating the design of an optimal relational database schema tailored specifically to the operational requirements of the WS Lounge.

3.6 System Architecture and Design

3.6.1 Context Diagram



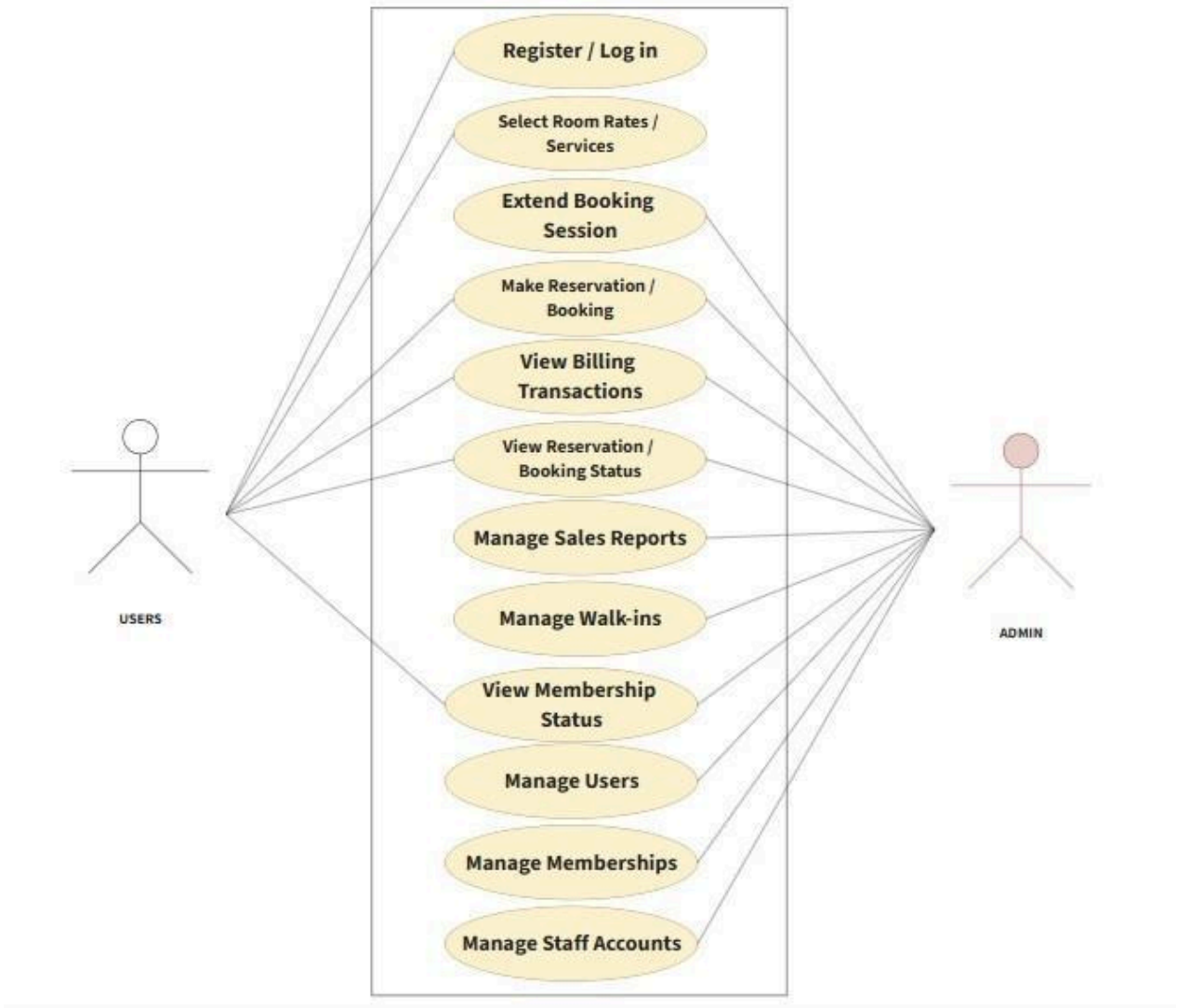
3.6.2 Data Flow Diagram



Significance of Context and Data Flow Diagram

The hierarchical structure of the Context Diagram and Data Flow Diagrams (DFDs) provided a detailed operational blueprint for the Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch. This structured approach systematically mapped the flow of data from the high-level conceptual overview down to granular database transactions, ensuring that every operational event—from initial room bookings to end-of-day sales updates—was accurately captured, validated, and logged.

3.7 Use-Case Diagram



Significance of Use-Case Diagram

The Use-Case Diagram for the Integrated Study Hub

Management System for WS Students & Professionals Lounge – La Paz Branch provided a structured behavioral model that outlined the precise interactions between external actors and the internal functional capabilities of the application. By defining a clear system boundary, this diagram served as an essential blueprint for verifying that all functional requirements mapped directly to specific user roles, eliminating structural conflicts prior to software deployment Sommerville [1]. The system architecture segregated operational responsibilities into two primary actor categories: Users (representing Students and Working Professionals) and the Admin (representing Front-desk Personnel and Executive Management) of the lounge.

From the Users' operational domain, the interface was engineered to offer an intuitive self-service portal that streamlined client interactions with the facility. Based on the system's behavioral logic, the functions explicitly made available to general clients included:

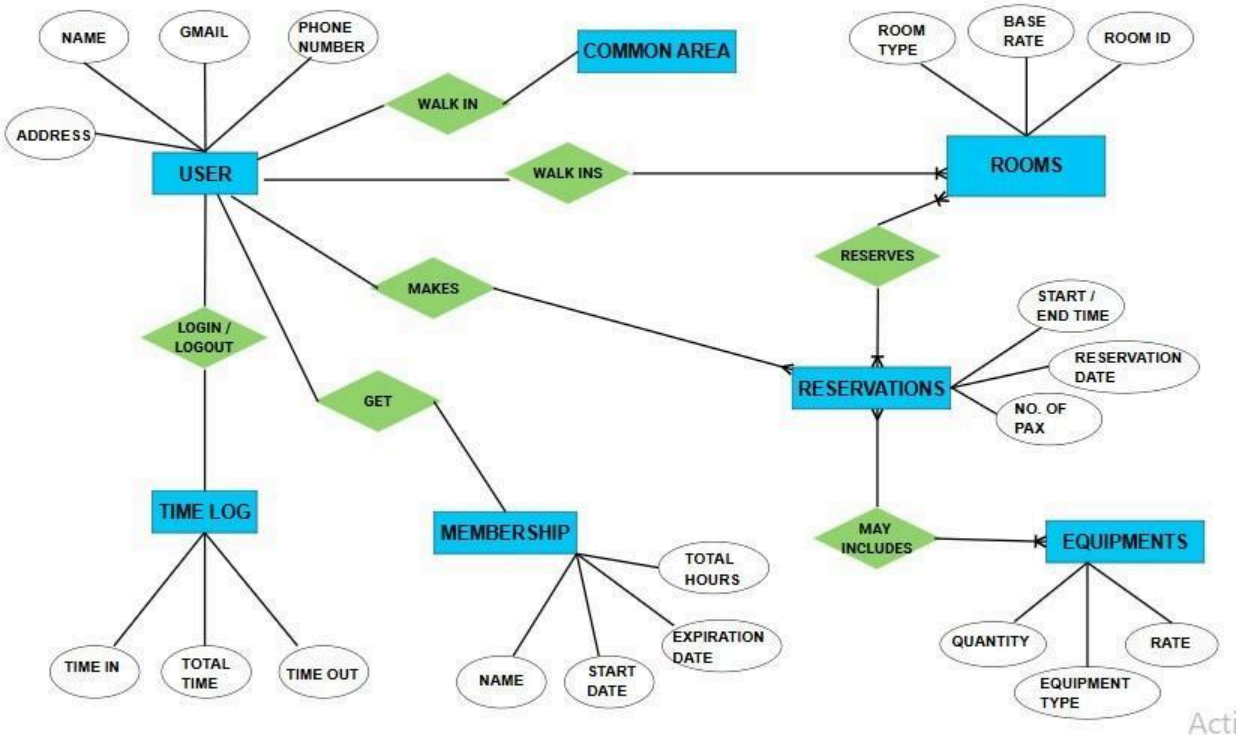
- Register / Log In – The gateway authentication module for securing user credentials and establishing authenticated sessions.
- Select Room Rates / Services – Allowed users to view realtime workspace availability, amenity options, and pricing matrices.
- Make Reservation / Booking – The primary functional control for securing a future schedule within the hub.
- Extend Booking Session – Enabled dynamic, real-time extensions of active room allocations depending on timeslot availability.
- View Reservation / Booking Status – Tracked real-time confirmation statuses of pending or active room requests.
- View Membership Status – Displayed active subscription details, remaining validity days, and tier perks.
- View Billing Transactions – Provided a transparent, itemized ledger of personal usage expenses and historical payments.

Conversely, the Admin actor possessed elevated operational permissions designed to govern daily logistical workflows, manage physical facilities, and maintain strict data governance over business assets. As illustrated by the multi-line connection schema, the Administrator could oversee user-level modules while holding exclusive privileges to execute the following administrative use cases:

- Manage Walk-Ins – Facilitated real-time check-ins, time-log initiations, and duration tracking for on-site clients without prior bookings.
- Manage Users – Granted administrative control over customer account profiles, restriction levels, and registration lists.
- Manage Memberships – Governed subscription package distributions, customer tier configurations, and expiration updates.
- Manage Sales Reports – Allowed management to aggregate financial data, track revenue streams, and generate analytical business summaries.
- Manage Staff Accounts – An exclusive backend administrative capability utilized to supervise employee system privileges, generate staff credentials, and enforce Role-Based Access Control (RBAC).

In synthesis, the use-case model validated the role-based architecture of the platform. By systematically organizing these lines of interaction, the system guaranteed that while clients enjoyed a frictionless, self-service workspace registration experience, hub administrators were equipped with the backend operational controls required to uphold financial transparency, system security, and high organizational efficiency.

3.8 Entity Relationship Diagram (ERD)



Significance of Entity Relationship Diagram

The Entitiy Relationship Diagram (ERD) served as the foundational architectural blueprint for the data management layer of the Integrated Study Hub Management System for WS Students & Professionals Lounge – La Paz Branch. By providing a visual and structural representation of the relational database schema, the ERD enabled the proponents to conceptualize and map how complex business operations—such as multi-tier room reservations, dynamic time-logging, membership subscription tracking, and itemized payments—were translated into normalized data entities Pressman & Maxim[2].

The primary significance of this diagram lay in its capacity to enforce structural data integrity and prevent relational anomalies across the application. Through the precise application of cardinality rules, primary key (PK) specifications, and foreign key (FK) constraints, the diagram mapped out the direct dependencies between core user activities, physical facility assets, financial logs, and security parameters, ensuring that orphan records were prevented during high-volume operational cycles.

As a formal database design map, each entity defined in the ERD correlated directly to physical relational database tables where attributes, data types, and key constraints were explicitly declared to maintain high data consistency and enable efficient query indexing Silberschatz et al[3]. Furthermore, the ERD served as a critical validation tool between the proponents and lounge stakeholders, verifying that the data persistence layer of the application aligned perfectly with the physical business rules, transaction pipelines, and administrative workflows of the lounge.

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CHAPTER IV

RESULTS AND DISCUSSION

The data gathered, analyzed, and interpreted during the development, implementation, and evaluation phases of the Integrated Study Hub Management System. It highlights the key findings and evaluates the system’s effectiveness in achieving its core objectives, specifically in automating transaction flows, implementing Role-Based Access Control (RBAC), and resolving the operational bottlenecks at the WS Students & Professionals Lounge.

The presented outcomes were derived from comprehensive system performance evaluations, User Acceptance Testing (UAT), and operational workflow assessments conducted during the system's deployment. Ultimately, this chapter demonstrates how the newly developed platform addressed the logistical vulnerabilities and administrative challenges identified in the earlier parts of the study, thereby validating the proponents' goal of establishing a streamlined, accurate, and efficient management solution for the study hub.

4.1 Task Manager

4.1.1 Gantt Chart / Project Timetable

Week	Task	Programmer	UI/UX Designer	Documenter

Week 1	Project Planning & Requiremen ts Gathering	Defined the technical stack and system requirement s.	Conducted user interviews and established the customer journey map.	Conducted user interviews; Created initial drafts of Chapters 1 and 3.
Week 2-3	Research & System Architectu re Planning	Finalized the backend framework and determined the database logic data flow.	Researched best practices for lounge booking interfaces and planned wireframes.	Completed the literature review for Chapter 2.
Week 2-3	Agile Framework Developmen t	Assisted in mapping out sprint cycles.	Assisted in defining user stories for the hub.	Created and structured the Agile Scrum Framework documentatio n.

Week 4-5	Use Case Diagram	Assisted in technical scoping.	Contributed to user role definitions.	Developed the system's Use Case Diagram.

Week 4-5	Data Flow Diagram	Mapped the data input/output routes.	Assisted in workflow visualization.	Formulated the system's Data Flow Diagram (DFD).
Week 4-5	Entity Relationship Diagram	Designed the Entity Relationship Diagram (ERD) for data optimization.	Formulated user interface schema dependencies .	Documented the ERD and detailed the specific data attributes.
Week 6-7	Database Design & Schema Creation	Developed the final database schema, including tables for Users, Reservations, Memberships, and Time Logs.	Reviewed data input field structures for the UI.	Documented the Database Design and created the necessary table specifications.
Week	UI	Reviewed	Developed	Assisted in

8-10	Wireframes and Mockups	technical feasibility of the layouts.	and finalized interactive mockups for the Landing Page, Dashboards, and Booking Page.	organizing visual design documentatio n.
Week 10-12	Frontend Developmen t	Began developing core system logic and API connection points.	Implemented approved UI mockups using HTML, CSS (Bootstrap), and JavaScript.	Documented the frontend architecture and user interface manual.
Week 12-16	Backend Developmen t	Coded backend system architectur e, including RBAC engines, reservation logics, and automated billing.	Refined components to match backend data responses.	Tracked development milestones and compiled technical notes.
Week	Integratio	Integrated	Coordinated	Documented

16-18	n of Frontend and Backend	backend APIs with the interactive frontend modules.	with the programmer to adjust responsive layouts.	integration test cases and data flows.
Week 18-19	System Security Testing & Debugging	Debugged backend logic and validated transaction processing data.	Patched responsive UI/UX flaws and navigation bugs.	Logged active test cases, system errors, and resolution results.
Week 20	System Evaluation	Optimized database response times and query speeds.	Implemented aesthetic refinements based on initial user feedback.	Summarized evaluation outcomes and final performance metrics.
Week 21-22	Documentat ion Finalizati on	Reviewed technical description s in the paper.	Finalized system figures, diagrams, and captions.	Compiled and completed the Final Research Documentatio n.
Week 23-24	Presentati on and Submission	Set up the backend environment for live demonstrati	Polished the client-side presentation interface.	Formulated the presentation structure and

		on.		finalized all project deliverables .

4.2 Presentation of Results

The development of the Integrated Study Hub Management System was driven by the necessity to transition WS Students & Professionals Lounge into a fully digitalized environment, optimizing how user interactions and workspace assets were administered. To validate its efficacy, the software underwent a series of technical evaluations that tested its resilience against common operational friction. The engineering of the platform emphasized data governance through a structured user hierarchy, ensuring that administrative privileges were securely partitioned.

During the operational trials, the application was subjected to stress testing to observe its stability in managing synchronous reservation inputs, logging actual client check-ins without database lag, and ensuring that time-dependent fee calculations remained accurate. Furthermore, a user-centric assessment was executed involving active front-desk operators, business managers, and daily clients. These participants navigated through the client portal, validated the automated billing adjustments for varying membership statuses, and audited the generation of financial history sheets. The empirical feedback gathered from these assessments demonstrated that the application completely eradicated the operational risks of overlapping schedules and manual calculation errors, successfully achieving the project's benchmark for institutional modernization.

4.3 Impact of Integrated Study Hub Management System of WS Students & Professionals Lounge on the Information Technology Field

The architecture of this project offered an architectural case study in the field of localized corporate automation and specialized database management. It confronted a prevalent challenge in small-scale hospitality and co-working sectors: the deployment of a centralized transaction framework that balanced client data integrity with dynamic, time-sensitive space allocation.

By engineering a solution that displaced fragmented administrative methods—such as paper-based logs, decentralized messaging systems, and manual timekeeping—this research provided a concrete methodology for small-to-medium enterprise (SME) software integration. It introduced a practical paradigm for multi-tier system layouts, illustrating how consumer-facing dashboards, real-time employee interfaces, and administrative reporting engines could co-exist within a single relational database. Consequently, this system elevated the regional benchmarks for operational technology, establishing that cloudready workflows could dramatically suppress human oversight errors while remaining aligned with the compliance directives of the Data Privacy Act of 2012 (RA 10173).

4.4 Contribution to Technological Advancement

The principal innovation of this system centered on its capability to align abstract digital scheduling with concrete physical facility tracking into an automated operational lifecycle. It introduced an automated bridge where a consumer's booking intent directly triggered computational time logs and algorithmic invoice creation.

By employing a customized role-based architecture, the software introduced a strict security boundary, ensuring that high-level administrative intelligence and overall business audits remained strictly confidential to executive accounts, while maintaining a transparent self-service interface for the general user base. Moreover, the algorithmic execution of automated pricing tiers within the subscription module demonstrated how business operations could be optimized through programmatic workflow controls. This system served as a functional model of how modern software methodologies could reengineer contemporary commercial workspaces, establishing a foundation for more agile, accountable, and secure transaction tracking systems.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Findings

The Integrated Study Hub Management System was conceptualized and developed to resolve the operational inefficiencies at the WS Students & Professionals Lounge – La Paz Branch. Prior to the study, the lounge heavily suffered from front-desk bottlenecks, data fragmentation, and service delays due to its reliance on non-integrated tools like physical handwritten logbooks, standalone spreadsheets, and Facebook Messenger. To bridge these operational gaps, the proponents aimed to build a unified, web-based platform that streamlined administrative workflows through interactive room reservations, automated billing, and Role-Based Access Control (RBAC). The development process utilized a systematic engineering approach, supplemented by extensive system testing and user evaluations to measure the platform's core functionalities. The collected data demonstrated that the system effectively minimized administrative workloads, eliminated resource double-bookings, and provided a highly secure environment for financial and customer records. Ultimately, the results validated that the developed system was robust, practical, and highly suitable for optimization in commercial study hub operations.

5.2 Conclusion

The successful implementation of the Integrated Study Hub Management System effectively resolved the primary administrative vulnerabilities encountered by the WS Students & Professionals Lounge – La Paz Branch. By replacing decentralized legacy methods with a database-driven architecture, the system

systematically addressed critical issues such as manual tracking errors, delayed response times, and disorganized data retrieval.

The inclusion of features like real-time workspace tracking, instant billing computation, and role-based access restrictions substantially elevated the lounge's operational speed, transactional accuracy, and data security.

The favorable feedback gathered from the system's end-users proved that the platform satisfied the precise demands of the study hub's staff and customers. In conclusion, the project underscored that integrating a centralized digital system was crucial for modernizing service-oriented businesses and achieving long-term operational excellence.

5.3 Recommendation

1. Conclude Continuous Staff Training – Provide ongoing orientation and quick-reference user guides for current and future frontline personnel to ensure smooth system navigation, especially during high-traffic peak hours.
2. Execute Preventive System Maintenance – Conduct periodic database optimization, security patch updates, and automated data backups to sustain the system's stability and safeguard transaction records.
3. Incorporate Next-Generation Features – For future iterations, developers may integrate digital payment gateways (like GCash and Maya) for cashless reservation processing and deploy automated SMS or email alerts for real-time customer booking updates.

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